Supplement to Coarse Marginal Resemblance and Model-Dependent Treatment-Response Uncertainty in a Fixed LLM Persona Panel

Yohei Nakajima
Untapped Capital, Bellevue, Washington, USA
Correspondence: yohei@untapped.vc
ORCID: https://orcid.org/0009-0009-1293-6053

This journal supplement preserves the complete uncertainty, secondary-result, and correction record underlying the shortened main article. Sections S1–S5 concern the fixed historical deployment; no new observations were collected. The separate Data and Instructions appendix gives exact prompt and data pointers. All historical evidence URLs below are pinned to commit 62ab90beff1d291b069c87f13b6b6b3495171ddd.

S1. Literature and scope

Positioning relative to prior work. Li and Ji (2026) establish using three LLMs and 59,508 participants from 62 countries that descriptive fit and intervention-effect accuracy can diverge, that prompt refinements improving realism do not reliably improve effect accuracy, and that errors vary with intervention logic, outcome structure, and attitude–behavior coupling. Persson, Schultzberg, and Ankargren (2026) formalize when LLM outcomes can serve as causal surrogates and why novel interventions still require human evidence. Lin et al. (2026) show that interventions can change the implicit simulated population even when explicit personas are fixed. Statistical-realism, persona-collapse, and state-versus-trait work further show that persona-conditioned populations can compress or misallocate heterogeneity (Xie et al. 2026; Harry et al. 2026; Xiao et al. 2026). Our differentiation is therefore not the broad divergence or the existence of mechanisms. It is a fixed-panel decomposition within a registered experimental program: the same explicit prompt panel is evaluated for composition, representation sensitivity and represented-continuation response, alongside an exploratory mean-band comparison. Original predicate families and later post-adjudication analyses retain separate evidentiary tiers, with exact prompt provenance and public inferential correction. The observed composition pattern is complementary to, not exclusive of, latent-user drift.

LLM strategic behavior. Akata et al. (2025) characterize repeated-game play modulated by prompts; Pal et al. (2026) elicit strategies from five models while varying continuation probability, payoffs, horizon knowledge, and framing; counterfactual-reasoning evaluations alter labels and payoff structures (Georgousis et al. 2026); and "strategic robustness" has been defined as payoff-preserving invariance across narratives (Mousavi Davoudi et al. 2026). These works establish that neither repeated games nor prompt/payoff perturbations are new. Our distinct combination is the fixed persona panel, the explicit between/within/response decomposition, exact prompt provenance, phase-specific registration records and mechanical adjudication followed by public inferential correction.

Strong positive evidence, and a different estimand. Ashokkumar, Hewitt, Ghezae, and Willer (2026) use study descriptions to forecast 469 effects from 70 preregistered, nationally representative survey experiments and find strong correlations with realized effects, alongside systematic effect-size overestimation and weaker performance in a megastudy archive. That is important contrary evidence against any blanket pessimism about LLMs in experimental science. It is also a forecasting task over studies rather than subject-level simulation of a response surface. Strong effect forecasting is compatible with the fixed-panel composition pattern studied here and reinforces the prescription to validate a simulator on the exact response object for which it will be used.

Synthetic participants and personas. Bisbee et al. (2024) find plausible survey averages alongside compressed variance, distorted coefficients, and temporal drift; Boelaert et al. (2025) report excess homogeneity; Anthis et al. (2025) catalog diversity and generalization challenges; Hullman et al. (2026) propose statistical calibration for confirmatory use; and Park et al. (2024) show that rich interview conditioning can substantially outperform lightweight demographic/persona descriptions. Format sensitivity (Sclar et al. 2024), role-play framing (Shanahan et al. 2023), persona collapse (Xiao et al. 2026), state blindness (Harry et al. 2026), and reviews of persona-experiment transparency (Batzner et al. 2025) all caution against treating a persona string as a stable human analogue. RLHF-related diversity reduction is a possible mechanism for concentrated policies, not a mechanism identified by this design. A fuller literature map and differentiation table are included in the public research record described in S4.

Technical context is retained explicitly: Berger et al. (2009) discuss reference-prior construction, not a guarantee that a chosen symmetric prior is harmless at six episodes; Efron and Tibshirani (1993) provide bootstrap foundations, not small-sample coverage for every percentile interval; Lehmann and Romano (2005) and Westfall and Young (1993) provide simultaneous-testing context, not retrospective registration. Clopper and Pearson (1934) concern binomial intervals. Hoeffding (1940) and Sklar (1959) concern dependence with given marginals; our variance/coupling applications are elementary. Holtzman et al. (2020) address decoding diversity, which is not a causal explanation of the present composition. Construct validity (Cronbach and Meehl, 1955), pattern-oriented modeling (Grimm et al., 2005), agent-based validation (Windrum et al., 2007), and policy-invariance concerns (Lucas, 1976) are methodological context. Assay sensitivity in Temple and Ellenberg (2000) and ICH E10 (2000) is an analogy, not a clinical equivalence test applied here.

Polachek et al. (2025) already compare stakes comparative statics in LLM PD behavior with a human study, as well as order/framing effects. This article adds the fixed-panel decomposition and correction record; neither behavioral-response comparison nor framing sensitivity is first established here. Kumagae and Kaneko (2026) study later survey respondent-profile coverage; that joint-profile target is distinct from represented continuation response.

S2. Chronology, provenance, and correction

The document header alone does not establish registration. Phase-specific evidence is unequal: Phase 4's release anchor at 2026-07-24T19:04:16Z precedes the earliest captured GPT response-created time 20:26:44Z; Phase 3 relies on internal adjudicator flags/study ordering without that external response-time evidence. Ordinary request event timestamps in the snapshot are blank; neither a complete request-time audit nor provider-signed chronology is claimed. Phase 3's claims registry and study rows are compared by the adjudicator; the report records all ten main claims and X1 with `postRegistered=false`, while disclosing that study-runner ordering preceded the later adjudicator-level flag. X1 was result-informed after Phase 3 results, registered before X1 data on July 24, 2026. Phase 4's registry seal is dated 2026-07-24T18:58:15Z; seal commit d24ee94ea722d817f0382519445c5e9d11a7bb1f was externally anchored by the release at 19:04:16Z. The tag was annotated, not GPG-signed; no signature is claimed. Its schedule and arm bytes were not altered by the seal.

Phase 4 A-OVH-1 amended overhead 900→1,000 after the earlier sentinel record but before its affected dispatch. A-OVH-2 corrected arithmetic to 1,250 before check 7/block E; check 6 had already run under A-OVH-1. These are result-informed prospective amendments, not wholly pre-study decisions. The X2-confirmation schedule omission is an additive mechanically derived schedule amendment. Phase 5 registry-v4 sealing preceded entry and dispatch on July 28. Amendment 1 corrected caps from 8,984 to 11,185 using sealed seeds, with zero Phase 5 calls at approval; it was pre-data and did not change designs or thresholds. The scope file's literal "PROPOSED — UNSEALED" header remained in sealed bytes, hash a0389d9f32

68d686ff2f4f8b93fa65c43aad77a35c52076ed1f365b4d935c0e6; the living scope-status addendum explains this. No journal edit changes that history.

Author-commissioned AI post-adjudication review began with the multiplicity, dependence, comparator, and construct criticisms. Round 2's methods reviewer later specified analyses and helped integrate them; this is disclosed assistance, not fully independent experimental replication. GitHub Actions executed fixed-seed analyses against archived data. Later reviewer-supplied anonymous-clone checks and outside capsule reproduction concern reconstruction; the archived Round 3/4 memos do not identify their AI tool/model, so no independent human certification is inferred. The favorable and unfavorable family analyses, prior sweeps, twelve refuted author predictions, and withdrawn p13 interpretation remain recoverable.

Exact chronology sources (all pinned):

- docs/phase3-preregistration.md
- docs/phase3-report.md
- docs/phase4/seal-record.md
- docs/phase4/budget-amendments.md
- docs/phase4/execution-schedule-amendments.json
- docs/phase4/provenance-notes.md
- docs/phase5/seal-record.md
- docs/phase5/amendment-1-caps.md
- docs/paper/scope-seal-status.md
- docs/reviews/README.md
- docs/reviews/round-2-methods-review.md
- docs/analysis/dead-predictions-final.md

S2.1. Research assistance, tool history, and relationships

Subject models generated the observations: primary requests named gpt-4.1 and archived responses named gpt-4.1-2025-04-14; Gemini named gemini-2.5-flash without an immutable deployment revision. A rejected Phase 4 entry candidate named claude-haiku-4-5 and returned claude-haiku-4-5-20251001. Its 16-token truncation failure is part of apparatus testing, not evidence that the manuscript-review tool used Haiku. Earlier exploratory Phase 2 code named GPT-5 Mini via Replit AI Integrations, without a recorded immutable model revision; it is outside this article's confirmatory boundary.

Research assistance was separate. Git commits dated July 24–29, 2026 attribute API/engine implementation, Phase 5 closeout, analysis integration and manuscript revisions to Replit Agent; its underlying foundation-model version is not recorded. The Round 2 methods review later expanded into specifying post-adjudication analyses and assisting integration; its model identity is not named. GitHub Actions executed fixed-seed archive analyses and builds; workflow execution and bot commit attribution do not identify a language model or establish independent human verification.

The public archive names Claude reviews of v7, v8, v9 and v13 and final precision corrections, archived July 29–August 1, 2026. It names Explore Science reviews of v6, v10, v11 and v12, archived July 29–31. These were author-side AI manuscript/artifact reviews that informed corrections, not independent journal peer review. The records preserve reviewed artifact hashes, criticisms and dispositions, but not exact Claude foundation-model revisions, Explore Science backend models/service versions, complete access logs, or every assistance session. Generic reviewer labels are not expanded into guessed model identities. Later reconstruction checks concern the archive, not human substitutability or validation of all scientific claims.

In November 2026, Codex assisted compression, literature/metadata checks, reconstruction of this disclosure, an isolated native-audit wrapper, and source/PDF packaging and checks. The task requested gpt-6.1-sol with high reasoning; this names the preparation configuration, not independently attested backend identity. No new subject-model observations were generated. The source-owned tool-use-reconstruction.json/.md records pinned paths, hashes, commit dates and recoverable roles; dates of archival commits need not be exact service-access times.

The author is Founder and General Partner of Untapped Capital, with a financial interest through that role. Untapped Capital supported this research, including computational resources and model/API usage, and publicly lists Explore Science as a portfolio company. The author is a developer of ActiveGraph, the event-sourced runtime used here. These relationships are disclosed rather than treating the support as independent third-party funding or the AI reviews as arm's-length journal review. The human contribution and responsibility statement appears in the main article. The archive documents selection of research questions, approval of registered designs, adjudication of review recommendations and acceptance of responsibility; it does not prove personal independent verification of every derivation or reference.

S3. Complete design, uncertainty, and principal results

The primary requested alias is gpt-4.1; captured responses report gpt-4.1-2025-04-14. The request itself was not an immutable snapshot pin. Outputs used 16-token outputs and a fixed minimal behavioral-subject prompt containing no game-theory vocabulary or reasoning scaffold. Temperature was 0.7 except in the registered Phase 5 sweep at 1.0 and 1.3. On the primary OpenAI-compatible path, `temperature` and `max_tokens=16` were explicitly supplied; the assembled prompt set `top_p=1.0`, which the adapter intentionally omitted from the wire at 1.0, while `presence_penalty`, `frequency_penalty`, and `logit_bias` were not supplied and therefore inherited provider defaults. No tools or native structured output were used.

Before Phase 5 data existed, the sixteen persona sentences were generated with a seeded `mulberry32` registry (seed 20260728) as the full cross of two age bands and three binary trait dimensions: agreeable/competitive, patient/impulsive, and risk-averse/risk-seeking. The preregistered leaning label was assigned from those generated traits—cooperative-leaning iff at least two of agreeable, patient, and risk-averse—yielding eight prompts in each leaning stratum by construction; no behavior was used to assign it. Names, ages, and occupations remained uncontrolled semantic components of the complete sealed sentences. Phase 5 prepends one such sentence to byte-identical task text. The cross-vendor Gemini tier is descriptive; the original Claude Haiku candidate failed a registered entry gate and was replaced under an archived amendment. Environment randomness is seeded; provider-side generation is not claimed to be seeded. The released records preserve prompts, completions, rounds and phase-specific decoding/provenance fields; Phase 3 lacks later response IDs/body hashes, and a complete request timestamp record is unavailable.

S3.0 Sequential architecture and registration status

Phases 1–2 developed and repaired the instrument: Phase 1 established the historical prototype, and Phase 2 added mechanical re-adjudication and enforcement after that prototype exposed analyst discretion. Neither is treated as prospective confirmation.

Confirmatory phase: Phase 3; **Primary role:** Bare GPT-4.1 repeated PD, framing, and RPS; **Independence unit used here:** complete episode; **Registration status:** reported internal pre-data flags; external

timing evidence incomplete

Confirmatory phase: Phase 4; **Primary role:** Representation robustness, X1/X2 wording, counterfactual labels/payoffs, continuation assays, adversaries, and sentinels; **Independence unit used here:** complete episode; **Registration status:** each block registered before its own data; X1 was sequentially registered after Phase 3 but before any X1 data

Confirmatory phase: Phase 5; **Primary role:** Sixteen sealed persona prompts crossed with Phase 4 instruments; descriptive Gemini tier; **Independence unit used here:** complete persona prompt, with episodes nested beneath it; **Registration status:** confirmatory predicates registered before Phase 5 data; later sensitivities are explicitly post-adjudication

The paper's main empirical decomposition is Phase 5, interpreted using representation results from Phases 3–4.

The full event store contains 5,505 completed runs, 54,276 round events, 108,552 seat-round decisions, and 36,251 archived provider-request events. The public confirmatory replay contract verifies 4,916 Phase 3–5 runs: 320 registered Phase 3/X1 LLM runs, 20 deterministic Phase 3 baselines, 2,864 Phase 4 runs, and 1,712 Phase 5 runs; three additional completed legacy entry/diagnostic runs are also replayed but are not counted as confirmatory. A separate transactional ledger records 30,530 Phase 4–5 calls, 13,141,675 input tokens, and 45,247 output tokens. The low output-token average (about 1.5 per call) is expected because valid actions were ordinarily one-token completions even though `max_tokens=16`; the ledger excludes earlier phases and must not be conflated with the full event-store request count. Machine-readable count definitions are included in the public record (S4).

Confirmatory claims were registered before the data that adjudicated them and were mechanically evaluated in a fixed vocabulary. The historical two-sided interiority rule used Clopper–Pearson bounds on seat-level round-one trials. Because two seats share an episode, the submission analysis treats the complete episode as the independence unit. For prompt i, condition d, and complete episode e, let $C_{ide} \in \{0, 1, 2\}$ be the number of cooperating seats, $A_{ide} = \mathbf{1}(C_{ide} \geq 1)$, and $B_{ide} = \mathbf{1}(C_{ide} = 2)$. The episode mean is then $Y_{ide} = (A_{ide} + B_{ide})/2 \in \{0, 0.5, 1\}$. Simultaneous Clopper–Pearson intervals are constructed for the two binary components and projected onto $E[Y]$. For six episodes, $\hat{p}_i(d) = \{\bar{A}_i(d) + \bar{B}_i(d)\}/2$. For each aggregate contrast, sixteen equally weighted prompt cells contribute 96 episodes per condition; pooled component counts estimate $\bar{p}(d) = 16^{-1} \sum_i p_i(d)$. Four component-condition intervals split the total error rate by Bonferroni. If $[L_d, U_d]$ is the projected condition-mean interval, the contrast interval is

$$[L_{.90} - U_{.10},\ U_{.90} - L_{.10}].$$

Under the pooled independent-binomial component working model, this union-bound construction has at least 95% simultaneous coverage. For heterogeneous fixed-prompt probabilities the component totals are Poisson-binomial; lower variance alone is not a proof of every binomial-tail coverage claim. We therefore report the exact construction within its stated working-model domain and distinguish the fixed-panel Dirichlet and cluster sensitivities, rather than claim unconditional finite-sample coverage under arbitrary heterogeneity. Its role is a conservative small-sample projection, not a claim that the sixteen prompt propensities are homogeneous. It does not assume seat independence and does not collapse to zero uncertainty when all recorded episodes agree. A fixed-panel symmetric-Dirichlet sensitivity assigns each prompt-cell outcome distribution a prior over $\{0, 0.5, 1\}$ and projects $E[Y] = 0.5q_{0.5} + q_1$. The primary sensitivity uses Dirichlet(0.5,0.5,0.5); post-adjudication prior sensitivity also reports symmetric $\alpha = 0.25$ and $\alpha = 1$. The percentile cluster bootstrap is retained as a separate sensitivity.

The hierarchy is deployment → explicit persona prompt → condition → episode → seat → round → provider request. Phase 5's confirmatory unit is the complete persona sentence; name, age, occupation, and traits are bundled semantic treatments. Registered claims attach to the conditional finite-panel estimand for these sixteen prompts. Claims about a wider persona generator are exploratory at $n = 16$. Pairing the same explicit prompt across conditions identifies a prompt-indexed contrast, not necessarily a stable latent person's treatment effect.

Protocol glossary. P3-A3 is the [.36,.63] band for pooled bare high-continuation {.75,.90} means only, not a Phase 5 persona-band registration. The Phase 5 registry anchor block and freeze-packet predicates contain no such extension; the later human-anchor scorecard marks its persona comparison exploratory. Original low-continuation anchors .190/.1783 differ. The initial bare J/F instrument and selected D1 presentation are distinct. `S2-absent` and `S2-present` are the registered repeated-game wording families. **Switch-bearing** means the span whose adjacent substitution produced the largest preregistered ladder gap and subsequently passed held-out confirmation; S2-present contains that replacement-and-reposition operation, while S2-absent contains the original sentence. `P3-A3` is the Phase 3 registered broad-reference cooperation claim, with band [0.36, 0.63]. `P5-1a`, historically called the **corner-mixture predicate**, is the registered support condition that fires when the interior fraction in the exact-bare-twin restricted set is below 0.10 under the frozen seat-level rule; it is not a general theorem about mixture structure. `P5-1b` is the registered between-persona dispersion comparison. `P5-2` pools registered conflict cells and classifies whether choices follow task text or persona-conditioned direction. `P5-3(a)` asks whether any persona × wording pair has both continuation-probability cells interior and a positive slope lower bound; `P5-3(b)` asks whether each persona lane rejects the bare configuration's dominated swap-cell option at a registered minimum rate. Historical verdict labels remain visible even where post-adjudication analyses change their scientific interpretation.

Phase 5 condition matrix. The 96 Tier-A persona–condition units are the full cross of sixteen prompts with six conditions:

Code: `rep-d10-s2a`; **Condition:** repeated PD, δ=.10, S2 absent; **Role in the paper:** repeated-game level, variance, and response

Code: `rep-d10-s2p`; **Condition:** repeated PD, δ=.10, S2 present; **Role in the paper:** repeated-game level, variance, and response

Code: `rep-d90-s2a`; **Condition:** repeated PD, δ=.90, S2 absent; **Role in the paper:** repeated-game level, variance, and response

Code: `rep-d90-s2p`; **Condition:** repeated PD, δ=.90, S2 present; **Role in the paper:** repeated-game level, variance, and response

Code: `os-swap`; **Condition:** one-shot canonical-payoff label swap; **Role in the paper:** semantic-label/payoff conflict

Code: `os-community`; **Condition:** one-shot Community framing; **Role in the paper:** near-interior framing anchor

The registered P5-1a denominator was restricted to persona cells whose exact recorded bare twin failed the same interiority gate. An outcome-blind exact-twin completion fixed that set as `rep-d90-s2a` and `os-swap`: sixteen personas in each condition, hence 32 units. The Community twin passed the bare gate; the other three repeated-game cells lacked exact bare twins and entered only the unrestricted 96-cell secondary.

The machinery's boundary is explicit:

The pipeline can enforce a registered predicate exactly; it cannot guarantee that the predicate represents a valid estimand, test family, or construct.

The historical "exact-episode" label refers to discrete component-binomial arithmetic assuming iid episodes within a prompt-condition; pooled uses additionally require the pooled working model. It is not a certification of arbitrary heterogeneous-prompt coverage. Simultaneous coverage is four components within one contrast, not both wording families or all prompt rows. The figure assets retain historical labels, interpreted within these limits.

S3 Continued: complete results

S3.1 Prompt composition and model-dependent response uncertainty

Across the four repeated-game cells, the fixed-panel Jeffreys $\alpha = 0.5$ sensitivity yields median between-prompt shares of 63.1%, 70.5%, 66.1%, and 66.6%, with 95% intervals [49.4%, 74.5%], [57.3%, 81.3%], [52.7%, 77.1%], and [52.7%, 77.7%]. The stronger claim that between-prompt variation exceeds one-half is not prior-robust. In a zero-call symmetric-prior sweep, $\alpha = 0.25$ produces medians of 74.8%–82.5% and $P(B/(B+W) > 0.5) > 0.999$ in every cell; Jeffreys $\alpha = 0.5$ produces medians of 63.1%–70.5% with probabilities 0.970–0.998; and $\alpha = 1$ produces medians of 47.1%–53.5% with probabilities 0.325–0.700. Thus the data support substantial prompt-indexed composition across the sweep, while "dominant between-prompt share" is prior-dependent.

For comparison, finite-opportunity plug-in shares are 85.5%, 96.1%, 88.8%, and 90.2%, with conditional episode-bootstrap 95% intervals [82.0%, 93.8%], [94.6%, 98.9%], [86.7%, 94.6%], and [87.9%, 95.5%]. The plug-in view conditions strongly on recorded boundary concentration and tends upward; symmetric-Dirichlet posteriors shrink six-episode corner cells toward the interior and tend downward. They are bracketing descriptions of different uncertainty assumptions, not interchangeable estimators. Figure S1 compares the two composition views across all four repeated-game cells.

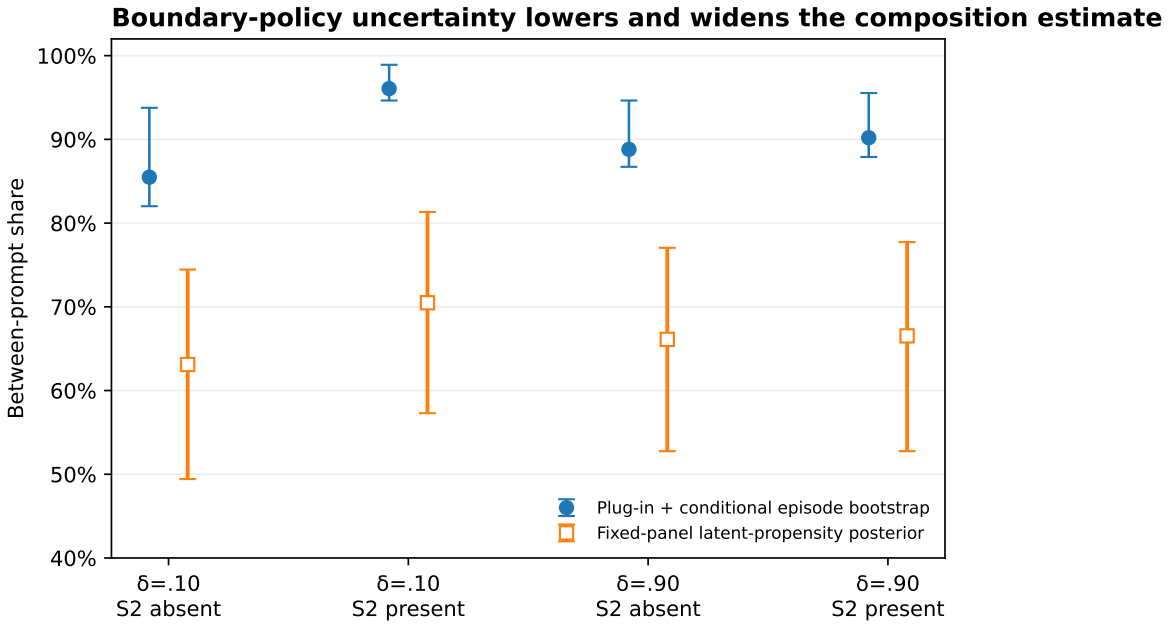


Figure S1. Between-prompt share of episode-level variation. The Jeffreys fixed-panel posterior propagates uncertainty in six-episode prompt cells; plug-in/conditional-bootstrap estimates describe the archived concentration more literally. A symmetric-prior sweep reported in text shows that the stronger "share above one-half" interpretation is prior-dependent.

Across the represented treatment, fixed-panel descriptive differences are +0.083 and +0.078. Pooled component-binomial working-model projection intervals are [−0.171,+0.330] and [−0.181,+0.330]; their coverage is conditional, not an unconditional result under arbitrary prompt heterogeneity. Four 98.75% binary component-condition intervals yield projected condition intervals of at least 97.5% and a contrast interval of at least 95% under that model. This applies to each contrast separately, not both contrasts or all 32 prompt rows.

The earlier design-effect heuristic DEFF=1+(m−1)s≈5.3–5.8 (m=6) treats prompt variation as a random generator contribution; its ≈16.5–18.2 episode-equivalent count is not the operative uncertainty constraint for the fixed sixteen prompts. For fixed independent prompt cells, variation of the sample mean instead depends on within-prompt episode variation; under simplifying common-variance conditions a factor near 1−s describes reduction relative to an iid marginal-population calculation. Neither heuristic supplies coverage or replaces an actual contrast analysis. Additional within-prompt dependence would affect both the contrast posterior/CP working model and P5-2 and is not eliminated by naming a cluster.

S3.1.1. New saved-data contrast sensitivity after two-model review

This analysis was specified after the two-model review, not preregistered. Its plan fixes seed 20261103, SHA256-derived label substreams, 50,000 draws at each α=.10,.25,.50,1 and 50,000 bootstrap draws per wording family, before execution. It uses only the frozen sixteen×four vectors of episode-category counts from episode-cluster-cells.csv; no subjects, provider calls, DB mutation or new experimental conditions. Exact plan/script/input/result hashes are in the candidate manifest; source files are contrast-sensitivity-plan.json, tools/contrast_sensitivity.py and contrast-sensitivity.json. An initial validation stopped before draws because displayed CSV means were rounded; only the equality tolerance was repaired, without changing seeds, draws or methods.

For each prompt×condition, posterior probabilities on {0,.5,1} are independent Dirichlet(counts+α); each draw averages sixteen latent means and subtracts low from high. This fixes the prompt panel and assumes independent episode/category likelihoods across prompt-cells. Central 95% credible intervals and posterior P(Δ>0) are:

α: 0.10; **S2 absent median [95%]:** 0.0793 [0.0285,0.1293]; **P(Δ>0):** 0.99854; **S2 present median [95%]:** 0.0745 [0.0299,0.1183]; **P(Δ>0):** 0.99918

α: 0.25; **S2 absent median [95%]:** 0.0742 [0.0140,0.1329]; **P(Δ>0):** 0.99162; **S2 present median [95%]:** 0.0694 [0.0135,0.1245]; **P(Δ>0):** 0.99156

α: 0.50; **S2 absent median [95%]:** 0.0666 [-0.0023,0.1340]; **P(Δ>0):** 0.97128; **S2 present median [95%]:** 0.0628 [-0.0038,0.1270]; **P(Δ>0):** 0.96766

α: 1.00; **S2 absent median [95%]:** 0.0554 [-0.0188,0.1292]; **P(Δ>0):** 0.92878; **S2 present median [95%]:** 0.0526 [-0.0204,0.1242]; **P(Δ>0):** 0.92196

The α=.10/.25 credible intervals exclude zero for both wording families; Jeffreys/uniform intervals include it. Shrinkage reduces the posterior mean contrast to 6/(6+3α) times the observed contrast, while latent corner uncertainty increases with α. Lower α favors simplex corners and is not universally less informative. These

are prior-conditional marginal credible intervals, not simultaneous confidence intervals or independent replications.

The two-stage scheme follows the existing bootstrap's prompt-then-episode resampling, extended to paired contrasts: sample sixteen prompt indices with replacement, use the same selected prompt at high/low, and independently sample six empirical episodes per condition/occurrence. This preserves prompt pairing but targets a hypothetical generator; the sixteen prompts were not randomly sampled from one. Its central 95% percentiles are [.0104,.1771] for S2 absent and [−.0104,.1875] for S2 present, medians .0781/.0729. Thus one excludes zero. Empirically unanimous cells remain point masses. No small-sample or design-based generator coverage is asserted. These marginal percentiles have no joint two-contrast multiplicity guarantee.

Independent analytic moment calculations verify the posterior and bootstrap Monte Carlo means/variances and frozen condition means; all planned outcomes are retained. Runtime was 1.37 CPU seconds with cached NumPy 2.4.6/SciPy 1.17.1. The CP intervals were independently reconstructed from component counts and match the archive. The contrast evidence is model-dependent; a broad imprecision headline based only on the widest construction would omit contrary sensitivities. All methods remain post-adjudication and do not establish matched-human effect accuracy, a uniquely correct mechanism, zero or equivalence.

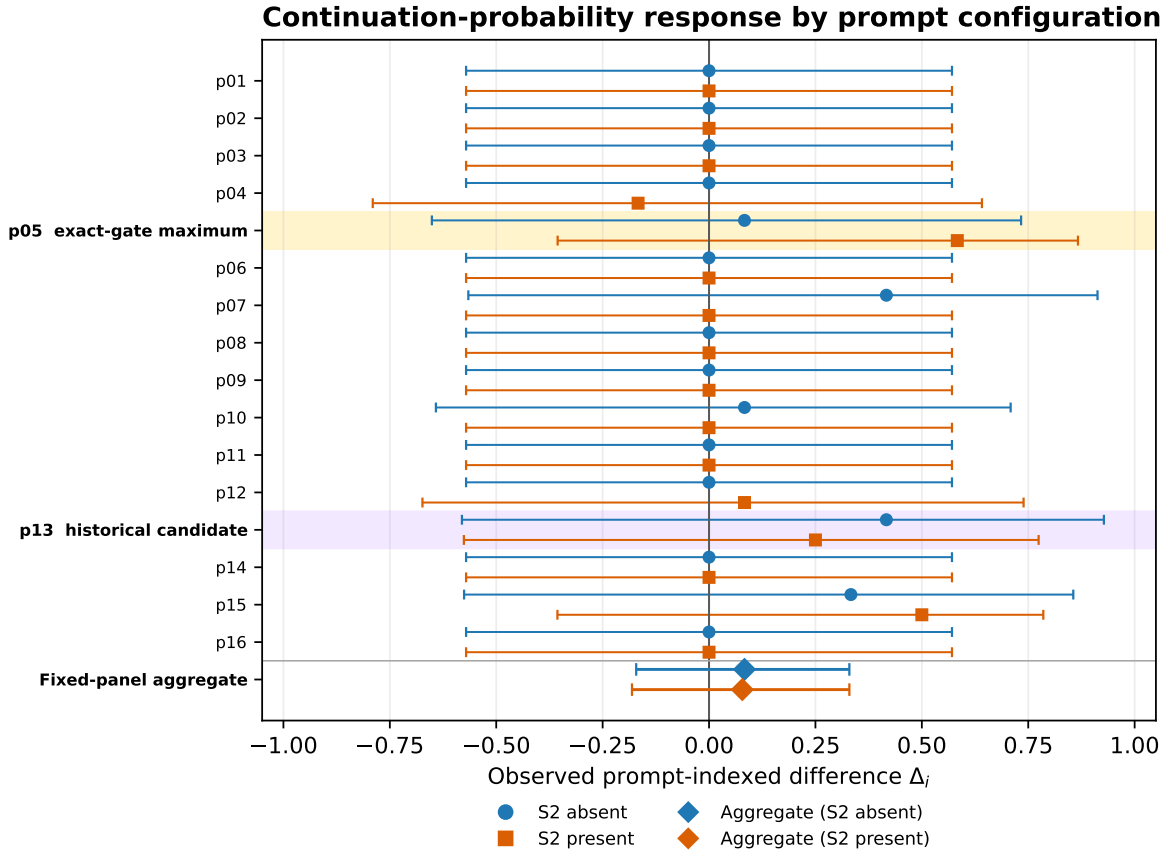


Figure S2. Prompt-indexed differences in round-one cooperation, $\Delta_i = \hat{p}_i(\delta = .90) - \hat{p}_i(\delta = .10)$, for both wording families. Bars are CP component projections assuming iid episodes within each prompt-condition: ≥95% per-row contrast coverage from four 98.75% components, not simultaneous over 32 rows; observed corners retain non-zero uncertainty. The p05 row is flagged as the largest candidate eligible under the component-binomial working-model family gate, and p13 as the historical candidate selected by the original rule.

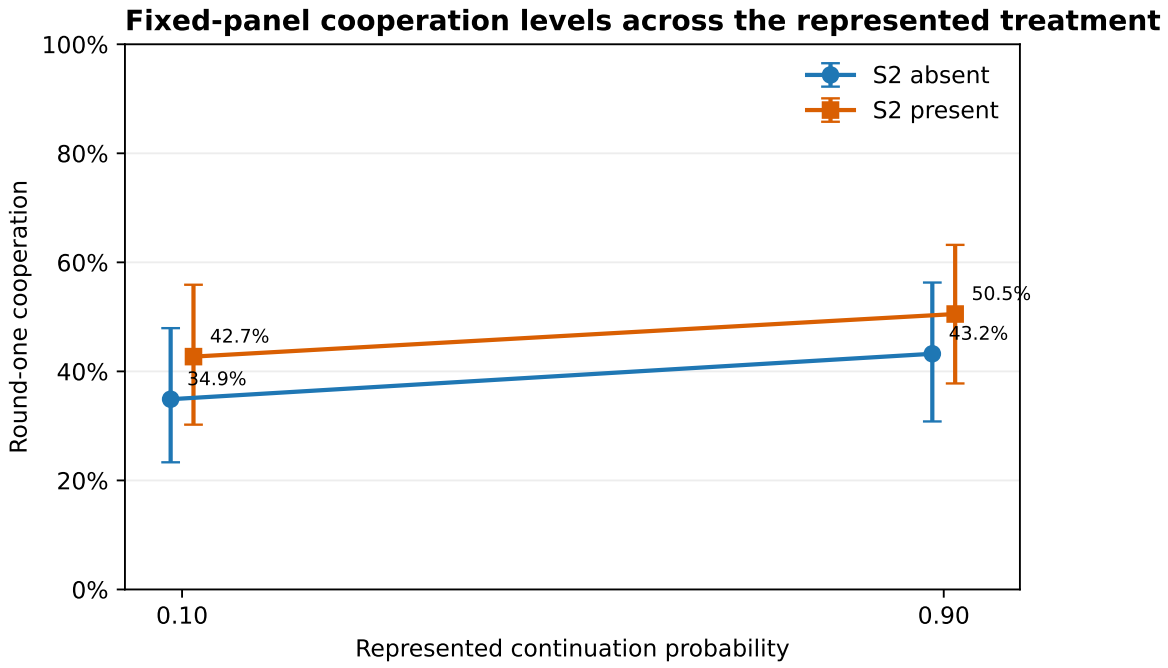


Figure S3. Fixed-panel cooperation by represented continuation-probability condition. Error bars are ≥97.5% condition projections under the pooled component-binomial working model, with ≥95% coverage separately for each contrast, not jointly across wording families; lines connect conditions for orientation only.

P3-A3's [0.36,0.63] band was registered for bare pooled high-continuation {.75,.90} means. The persona high-continuation means .432/.505 fall inside it, while low-continuation means .349/.427 exceed original low-condition human anchors .190/.1783. The historical three-of-four scorecard extends the band across conditions exploratorily; no Phase 5 band-extension predicate was sealed. These nonmatched human comparisons are descriptive. The preregistered leaning rule divides the panel into eight cooperative-leaning and eight defect-leaning complete prompts. Descriptive gaps range from 0.510 to 0.719 across the repeated-game conditions and are fixed-panel prompt-bundle contrasts, not causal trait effects:

condition: `rep-d10-s2a`; **cooperative-leaning mean:** 0.615; **defect-leaning mean:** 0.083; **difference:** +0.531; **prompts per stratum:** 8

condition: `rep-d10-s2p`; **cooperative-leaning mean:** 0.760; **defect-leaning mean:** 0.094; **difference:** +0.667; **prompts per stratum:** 8

condition: `rep-d90-s2a`; **cooperative-leaning mean:** 0.688; **defect-leaning mean:** 0.177; **difference:** +0.510; **prompts per stratum:** 8

condition: `rep-d90-s2p`; **cooperative-leaning mean:** 0.865; **defect-leaning mean:** 0.146; **difference:** +0.719; **prompts per stratum:** 8

condition: `os-community`; **cooperative-leaning mean:** 0.688; **defect-leaning mean:** 0.019; **difference:** +0.669; **prompts per stratum:** 8

P5-1b used protocol-nonmatched human SD references mechanically implied from Dal Bó and Fréchette's (2011) R=40 strategy-frequency estimates: 0.4122 for their $\delta = .50$ panel and 0.3116 for $\delta = .75$. The frozen ratio $\rho = .75$ gave thresholds 0.3092 and 0.2337. It was a preregistered heuristic tolerance, not an equivalence margin. Finite-opportunity-corrected plug-in SDs were 0.4182, 0.4784, 0.4408, and 0.4323. A

conditional episode bootstrap produced corrected-SD intervals [0.4122, 0.4391], [0.4696, 0.4916], [0.4279, 0.4654], and [0.4269, 0.4496], but unanimous cells remain point masses in that bootstrap. The first lower bound was independently reproduced at full precision; its displayed equality to the 0.4122 human reference is a rounding coincidence. A two-stage prompt+episode bootstrap changes the estimand toward a hypothetical persona generator and yields wider SD intervals [0.2724, 0.4879], [0.3696, 0.5123], [0.3457, 0.4890], and [0.3345, 0.4847]. P5-1b is therefore retained as a permissive historical checkpoint, not evidence of human-variance equivalence.

Across all six Phase 5 conditions, the historical seat-level rule classifies 14/96 persona-condition cells interior, the component-binomial episode projection 11/96, and the Jeffreys sensitivity 19/96. In the registered 32-unit set, P5-1a requires interior fraction <.10: frozen 3/32=.09375 supports it; post-adjudication projection 2/32=.0625 also satisfies it; Jeffreys 5/32=.15625 does not. The Jeffreys sensitivity therefore reverses the predicate; the frozen verdict stays historical. At $n = 6$ with a three-valued outcome, modest differences in interval width move cells across the threshold; continuous uncertainty and variance components are more informative than the binary census.

Descriptively, the composition pattern did not transfer to the endpoint-nonstationary Gemini tier: 9/24 Gemini cells met the historical interiority rule versus 14/96 in the primary panel, and several representation effects reversed direction. This is consistent with deployment-specific composition; the tier is not a formal replication. Human references remain protocol-nonmatched because continuation probabilities, payoffs, incentives, assignment, and experience differ from the present design; no matched magnitude or human-equivalence claim is made.

S3.2 Representation interventions expose prompt-surface control

For the bare configuration, round-one cooperation was 0.000 at every registered continuation probability. X2 decomposed the v1 and v2a prompt bundles into six sentence/block spans and constructed forward and reverse ladders by replacing one complete span at a time. The selected S2 operation replaced and repositioned "After every round there is a {deltaPct}% chance the session continues with another round" with "At the end of each round there is a {deltaPct}% chance that the session goes on for one more round." Screening used ten episodes per rung; the selected minimal pair was confirmed at temperature 0.7 on 20 fresh episodes per side (seeds 2953–2972), moving cooperation from 0/40 to 37/40. Because wording and position were one atomic operation, the design does not separate them.

The effect was specific to the representation of repeated interaction, not a generic main effect of wording. In the 640-episode one-shot D1 battery, GPT-4.1's registered wording main effect was +0.0063 (SE 0.0210; Holm-adjusted $p = 1.00$), and none of the registered wording interactions was supported. Conversely, the D1 presentation selected for the repeated assay had a one-shot mean of 0.100; when embedded in the repeated-game protocol, round-one cooperation rose to 0.750 at $\delta = .10$ and 1.000 at $\delta = .90$. The Community wrapper produced 1.000 in both conditions. These repeated cells were correctly classified as ceiling-confounded, so they do not identify a continuation-probability slope. They do show that announcing and implementing repeated interaction can be a much larger treatment than ordinary one-shot wording variation.

In the label-swap cell, canonical payoffs were held fixed while "Cooperate" and "Defect" were attached to opposite strategic roles. The bare configuration chose the cooperation-worded option 0/40 times and instead took the payoff-dominated role whenever it carried the word "Defect." This shows that the displayed label or a label-linked learned prior can override payoff dominance in this registered cell. It does not identify intrinsic lexical valence: a learned game-theoretic association such as "Defect = equilibrium/dominant action in

Prisoner's Dilemma" is an equally plausible mechanism. A structurally equivalent non-PD control retaining the same labels was not run.

Persona conditioning produces two observed contrasts, but they are not factorially separable. Differences among complete persona prompts generate the leaning gaps reported in S3.1. Adding any tested persona string reverses the bare swap-cell choice, yet no non-semantic prefix matched for length, punctuation, and position was run; semantic persona content cannot be isolated from generic sequence-format disruption. In the swap cell, label and payoff also point to the same option for persona-conditioned configurations, leaving the reversal mechanism ambiguous.

P5-3(b)'s 24 evaluable lanes comprise sixteen personas at T=0.7 plus p02, p06, p11, and p15 at each of T=1.0 and T=1.3. Under the iid prompt-cell component-binomial model, the one-sided Bonferroni projection across these 24 lanes retains all lower bounds above the frozen 0.20 threshold; the minimum is 0.462. The pooled P5-2 task-consistent share is 90/704 seat decisions across 352 episodes, equivalently 45/352=0.128 on episode means. The frozen historical adjudication used the seat-level rule. A pooled component-binomial projection gives [0.092,0.172] under the same working-model caution as aggregate Δ: it does not certify coverage under arbitrary heterogeneous probabilities or additional within-prompt dependence. A stratified bootstrap resampling forty persona×conflict-cell clusters gives [0.071,0.189]; this changes the conditioning toward resampled clusters and is not an automatic correction for the fixed-panel estimand or all dependence between cells sharing a persona.

Independent fixed-panel Dirichlet aggregation is materially prior-sensitive because every sparse cluster receives its own symmetric prior. Under the Jeffreys specification, each cluster receives total prior concentration 1.5, or 60 category-count units across forty clusters; this is not literally equivalent to sixty pooled episodes because the clusters have unequal sizes, but it is non-negligible relative to 352 observations and pulls the aggregate toward 0.5. A zero-call sensitivity sweep gives:

symmetric α: 0.10; **posterior median:** 0.138; **95% interval:** [0.124, 0.153]; $P(\theta \leq .20)$**:** 1.000

symmetric α: 0.25; **posterior median:** 0.152; **95% interval:** [0.135, 0.171]; $P(\theta \leq .20)$**:** >0.999

symmetric α: 0.50; **posterior median:** 0.172; **95% interval:** [0.152, 0.195]; $P(\theta \leq .20)$**:** 0.991

symmetric α: 1.00; **posterior median:** 0.205; **95% interval:** [0.182, 0.231]; $P(\theta \leq .20)$**:** 0.329

The Jeffreys posterior's proximity to the registered 0.20 boundary is therefore prior-dependent rather than an independent signal from the data; the $\alpha = 1$ posterior crosses the boundary. The cluster bootstrap and fixed-panel posterior answer different conditioning/target questions; neither supplies assumption-free dependence correction. Every repeated conflict subcell is mixed; only the swap cell is individually persona-dominant. The historical pooled classification is therefore mechanism-confounded and carried by the swap cell, not evidence of a general persona-dominance mechanism. Figure S4 summarizes the two bare-configuration representation interventions.

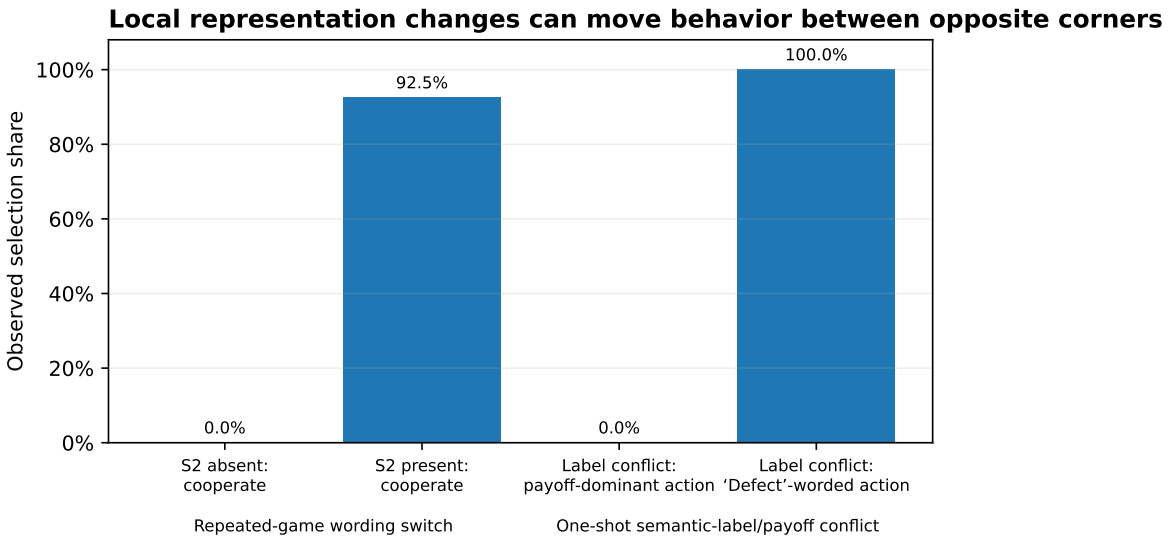


Figure S4. Two representation interventions in the bare configuration. The S2 wording-and-position operation moved cooperation from 0/40 to 37/40. In the one-shot label conflict, the payoff-dominant action was never chosen when the dominated role carried "Defect." These bars report selection shares, not a common or uniquely identified causal mechanism.

S3.3 What the marginal checks cannot identify

Let $p_i(d) = E[Y \mid i, d]$, where i indexes the complete explicit persona prompt and d the experimental condition.

Proposition A: broad bands only partially identify the aggregate contrast. If a synthetic condition mean is accepted within tolerance ϵ_d of a reference mean in each condition, then

$$|\Delta^S - \Delta^H| \leq \epsilon_0 + \epsilon_1.$$

Equivalently, accepted bands $[\ell_0, u_0]$ and $[\ell_1, u_1]$ imply only

$$\Delta^S \in [\ell_1 - u_0, u_1 - \ell_0].$$

Exact condition-specific mean matching would force the aggregate effect by the identity $\Delta = \mu_1 - \mu_0$; the logical slack is in coarse criteria. This is not a retrospective registered persona-band validation test; only the high-continuation bare band was originally registered. Applying [.36,.63] to both conditions would allow $\Delta \in [-.27, .27]$, illustrating tolerance slack rather than a new empirical predicate.

Proposition B: aggregate moments do not identify microstructure or response coupling. This is an application of the law of total variance and classical Fréchet–Hoeffding/Sklar coupling results (Hoeffding 1940; Sklar 1959) to synthetic-participant validation, not a new probability theorem. Mean and total variance do not identify how variation is divided between prompt configurations and repeated draws, nor do they identify distributional shape or boundary concentration. Even exact condition-specific distributions do not identify the cross-condition coupling and therefore do not identify the distribution of prompt-indexed responses $\Delta_i = p_i(1) - p_i(0)$. Reusing an explicit persona string supplies one prompt-indexed coupling, but

interpreting it as a stable synthetic individual's potential-outcome contrast requires latent-person invariance, which this study does not test.

The study therefore identifies a composition pattern in one fixed prompt panel. It does not establish that humans have a different microstructure, that RLHF caused the pattern, that latent-user drift is absent, or that the same pattern occurs across persona generators.

S3.4 The favored persona-level result is not prospectively confirmed; the archived family is underpowered

Under the historical seat-level rule, persona p13 moved from 0.333 cooperation at δ=.10 to 0.750 at δ=.90 and passed a per-candidate lower-bound test. The rule searched multiple persona × wording candidates and fired on any pass without declared family-level error control. Author-commissioned AI review identified that defect.

Three 200,000-permutation gate constructions are now reported; all permutation p-values use the add-one convention, $\widehat{p} = (r+1)/(B+1)$, with exact Monte Carlo intervals. Under the historical seat-level gate, p13 remains the maximum at +0.4167, with familywise $p = 0.059230$, Monte Carlo 95% interval [0.058194, 0.060268]. Under the percentile episode-cluster-bootstrap sensitivity, p13 also remains the maximum and $p = 0.043455$, interval [0.042561, 0.044353]. The episode projection has finite-sample coverage within a prompt-cell under iid episode draws and the component-binomial model; this is not pooled-heterogeneity coverage, nor simultaneous coverage over the candidate family; the percentile bootstrap is retained symmetrically but has no comparable small-sample coverage guarantee. Under the component-binomial working-model gate, p13 is ineligible: its low-δ lower bound falls below 0.05 and its high-δ upper bound exceeds 0.95. Only p04/s2p and p05/s2a pass both gates; the largest eligible slope belongs to p05/s2a (+0.0833), with familywise $p = 0.773206$, interval [0.771363, 0.775039].

For p13/s2a, the percentile bootstrap admitted both conditions as interior—δ=.10: [0.083, 0.667]; δ=.90: [0.583, 0.917]—whereas the component-binomial working-model projection rejected both—[0.047, 0.800] and [0.287, 0.954], respectively. Neither recorded cell was at an exact corner. The eligibility difference therefore arises from small-sample interval width and coverage behavior, not from a corner interval falsely passing the gate.

The complete data-dependent gate is dynamically reapplied within every permutation, not frozen from the observed-data mask. The implementation precomputes 56 possible-composition gate values and performs 25,600,000 condition-gate lookup applications at B=200,000. In a deliberately incorrect comparison that froze the observed-data mask, the maximum statistic differed from the dynamic procedure in 718 of 5,000 null draws (14.4%), showing that reapplication materially changes the reference distribution. Lookup/direct parity and the frozen-mask regression are included in the public audit record (S4).

An exhaustive attainability audit shows that, with six episodes per condition, 12 of the 28 possible episode-value compositions pass the working-model gate. Their sample means range from 0.333 to 0.667, but eligibility depends on the full $0, 0.5, 1$ composition rather than on the mean alone. Two eligible cells can therefore differ by at most 0.333. Under the archived 32-candidate null structure, the add-one familywise tail probability at that maximum attainable slope is 0.075040, with Monte Carlo 95% interval [0.073884, 0.076198]; no exact-gate result in this archived family can reach 0.05. This working-model analysis is therefore a valid dependence-aware sensitivity but not a powered disconfirmation of a p13-sized capability claim. The record neither prospectively confirms nor decisively disconfirms p13; it identifies a replication target whose next test must be sized prospectively (S5.3). A separately authorized prospective test could preregister the candidate family, episode-level dependence unit, interiority gate, maximum statistic, familywise decision rule,

and sample size before any data are collected.

These variants were specified and executed together only after author-commissioned AI review exposed the original defects. No familywise gate was registered at the original freeze, and the repository contains no seal-before-compute record for these sensitivities. The analyses ran in GitHub Actions against archived databases with fixed seeds, and their outputs were committed regardless of direction. Accordingly, the favorable $p = 0.043455$ variant cannot create prospective confirmation, just as the other variants cannot retroactively strengthen the frozen claim. The historical mechanical P5-3 verdict remains visible because clause (b) fired strongly and p13 passed the rule as frozen. Clause (a) did not prospectively establish the capability claim because the frozen search lacked family control; the working-model post-adjudication procedure is too underpowered at n=6 to provide decisive evidence against it. Figure S5 compares the three post-adjudication familywise constructions and their different eligibility rules.

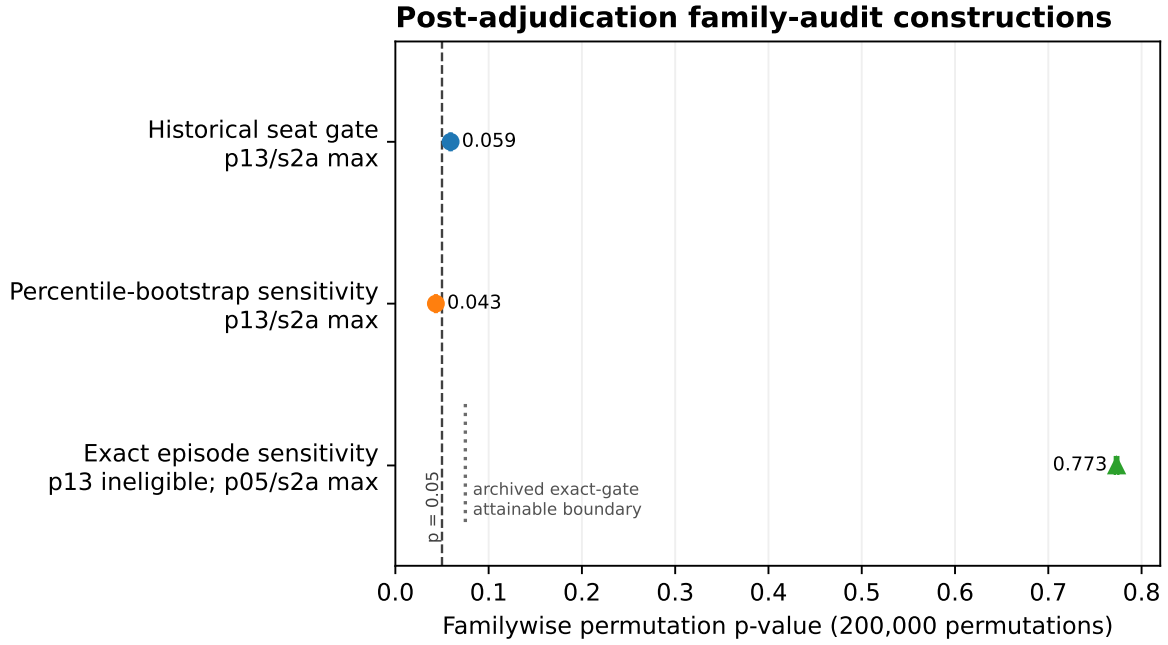


Figure S5. Post-adjudication familywise constructions. The first two points are p13/s2a under the historical and percentile-bootstrap gates. Under the component-binomial working-model gate, p13 is ineligible; the third point is the familywise result for the largest eligible candidate, p05/s2a. The dotted line at $p = 0.075040$ marks the estimated minimum attainable familywise p-value for the archived $n = 6$, 32-candidate working-model-gate design and applies only to that construction. None of the procedures was registered at the original freeze; see S3.4 for the bounded scientific status.

S3.5 Auditability

Twelve registered author predictions were refuted by data and published. Four underspecified analysis choices were resolved outcome-blind. A two-sided gate blocked a false ceiling conclusion. Sentinels detected a time-indexed behavioral discontinuity in an unversioned endpoint and exposed a monitoring gap that was repaired with an attestation gate. The public capsule now verifies all 4,916 confirmatory Phase 3–5 runs with zero live model calls: 4,896 LLM runs replay byte-exact and 20 deterministic baselines are independently recomputed. Author-commissioned AI review then found the family-error and dependence defects discussed above; the archived record made both diagnosable and correctable.

The claim is bounded: the machinery is procedurally exact and extensively auditable. Its strongest achievement is not self-validation but preservation of enough provenance for outsiders to identify where procedural correctness stopped short of statistical validity.

S4. Data availability and replay limits

The public repository contains the event stores, prompt registries, sealed registrations, adjudication records, timestamp proofs, post-adjudication analyses, figures, manuscript history, and review record. The one-command capsule verifies all 4,916 confirmatory Phase 3–5 runs with zero credentials and zero live model calls. The audit covers 320 registered Phase 3/X1 LLM runs, 20 deterministic Phase 3 baselines, 2,864 Phase 4 runs, and 1,712 Phase 5 runs; three additional completed legacy entry/diagnostic runs are also replayed but are not counted as confirmatory. Phase 3 replay re-renders every prompt, requires recorded-cache hash hits, reparses raw completions, recomputes actions, payoffs, and RNG draw counts, and checks recorded call parity. The deterministic P3-C3 baseline is independently recomputed from archived seeds and game objects.

Phase 3 used a legacy provider path without Phase 4–5 response IDs or deterministic request-body SHA capture. Phase 4–5 contain 30,421 normal request events and 30,397 response events; the 24-event difference is the disclosed provider-failure partial set. Each partial belongs to a failed run, was excluded from completed-run analyses and the 4,916-run replay denominator, and was never decoded as an action. Request attempts remain represented in request and budget accounting rather than being silently discarded; only completed replacement runs, where present under the registered replacement procedure, enter analysis. Those response records contain rendered prompts, bundle and request-body hashes, engine commit and provider route, raw text, and provider response IDs. Individual completion payloads were not provider-attested or separately hash-chained at receipt. Capsule checksum manifests and external timestamps make the released database snapshot tamper-evident relative to publication; replay cannot prove that no alteration occurred before snapshot sealing.

The immutable study checkout supplies the native capsule route (which uses a temporary loopback transport):

```
git clone https://github.com/yoheinakajima/synthetic-players
cd synthetic-players
git checkout 62ab90beff1d291b069c87f13b6b6b3495171ddd
cd capsule
bash verify.sh
```

All post-adjudication analyses are zero-call scripts over the archived databases and are labeled separately from prospectively registered results. The principal machine-readable surfaces are the Submission Analyses directory (count reconciliation, dependence sensitivities, family audits, and the final prior sweep), the Persona Table, the Literature Map and Novelty Relationships documents, and the Review Archive. Stable links to each are provided from the repository's review entry point, keeping the empirical argument readable without embedding long filesystem paths in the paper.

The prepared evidence/code ancestor 1bc9cb8dcb44cf514fe2fad53cdbfd96725badd0 supplies the offline wrapper and new contrast plan/script/results: https://github.com/yoheinakajima/synthetic-players/tree/1bc9cb8dcb44cf514fe2fad53cdbfd96725badd0/journal/jesa. The wrapper is absent from the immutable study commit; the final editorial head is separately identified in the candidate manifest/PR. Check out that ancestor (or this exact journal candidate) before following the separate Data and Instructions §3 route. Journal preparation used the native replay audit functions in process, avoiding the historical HTTP transport. The original

190-file capsule manifest passed before and after. The receipt and reports are supplied under `evidence/`; this was an offline recheck, not a new model run. Python 3.11.15 differs from the historical environment note 3.11.14. Both identities remain documented. The README line that ends at 4,916 omits three legacy diagnostics; the verifier covers 4,919 (4,899 LLM plus twenty baselines), including 4,916 confirmatory. Earlier 4,576-run hierarchy/review descriptions refer to an earlier boundary.

The actual repository LICENSE contains MIT terms for code and a CC BY4.0 declaration for released research data/reports/generated papers; CC BY4.0 gives the license terms. These pointers are repository declarations, not independent proof of all third-party/provider rights. The current Git study and preparation artifacts are identified by commit/hash; no completed trusted replication-archive deposit, additional permanent identifier or Data Editor validation is claimed. The applicability and exact form of those journal steps remain to be resolved against current authoritative policy.

S5. Retained secondary results and optional future work

S5.1 Temperature secondary

Choice entropy is defined as base-2 Shannon entropy, $H = -\sum_a p(a) \log_2 p(a)$, over round-one payoff-role choices. The historical registered secondary pooled all valid choices at each temperature, but the T=0.7 and higher-temperature samples had different composition. The matched-sweep reanalysis uses only persona-cell lanes observed at all three temperatures:

temperature: 0.7; **matched units:** 13; **seats:** 544; **pooled Shannon entropy (bits):** 0.8310; **mean within-unit entropy (bits):** 0.4484

temperature: 1.0; **matched units:** 13; **seats:** 284; **pooled Shannon entropy (bits):** 0.7822; **mean within-unit entropy (bits):** 0.2566

temperature: 1.3; **matched units:** 13; **seats:** 284; **pooled Shannon entropy (bits):** 0.7698; **mean within-unit entropy (bits):** 0.2877

The registered pooled decline is partly composition-confounded but survives on the identical sweep lattice. Mean within-unit entropy is reported separately because pooled entropy can remain high when different prompt-cell units occupy opposite boundaries. Neither statistic identifies a temperature mechanism. The plug-in Shannon estimates use unequal seat counts (544 versus 284 and 284), so their small finite-sample biases are not identical; no bias-corrected entropy claim is made.

S5.2 Other supplementary findings

The following registered or explicitly descriptive results are outside the paper's main composition–response arc but are retained here so that the completed experimental record is citable.

Opponent-contingent adversarial play. In the 50-round RPS adversary suite, a second-order n-gram opponent earned +0.215 payoff units per round against GPT-4.1 and survived Holm correction. A first-order tracker instead earned -0.118, meaning the subject beat it. The WSLS-targeter, designed around the earlier near-deterministic lose-shift signature, earned only +0.008 with a lower confidence bound below zero, although it outperformed the first-order tracker by +0.126. The shuffled-history control was not worse than the ordered tracker. Thus a behavioral signature measured against one opponent did not transport as a simple profitable rule against another; exploitable structure was opponent-contingent rather than captured by a single

global label. The GPT-4.1 opponent tests were registered; the six secondary arm tests were Holm-controlled, while the cross-opponent transport interpretation here is descriptive.

Role-attached asymmetry in a symmetric game. After RPS moves were renamed with neutral symbols and display order was exactly counterbalanced, the registered GPT-4.1 first-minus-rock contrast was -0.181: the preregistered positive-direction hypothesis was not supported, and the observed direction was reversed. A support-only Dirichlet analysis assigned probability 0.0001 to first-only selection exceeding rock-only selection. The cross-vendor mirror had the opposite sign, +0.243, with a one-sided lower bound of +0.139. The GPT-4.1 contrast was registered; the Gemini mirror is reported descriptively under the paper's cross-vendor tier. The pattern is descriptive evidence of vendor-specific, role-attached asymmetry in a formally symmetric game, not a universal "rock bias."

Cross-vendor label–payoff dissociation. In the canonical label-swap cell, GPT-4.1 chose the cooperation role 1.000 of the time while choosing the displayed word COOPERATE 0.000, thereby following the displayed token DEFECT even when it named the payoff-dominated role. Gemini chose the cooperation role 0.213 and the word COOPERATE 0.788, mostly following the payoff-dominant role instead. When counterfactual payoffs made the cooperation role strictly dominant and displayed it as DEFECT, the two models' cooperation-role shares were 1.000 and 0.975. For GPT-4.1, word and payoff dominance were congruent in this cell, so it does not separate the two channels. For Gemini, the cell is informative because the model moved to the payoff-dominant cooperation role despite that role carrying DEFECT. The GPT-4.1 cells were registered; the Gemini figures are descriptive under the paper's cross-vendor tier. The vendors differed in when a familiar action word overrode the supplied payoff structure; the data do not identify one universal lexical mechanism.

Endpoint drift and subject eligibility. The Gemini sentinel fingerprint changed from 10/10 baseline matches to a sequence reaching 6/10 and 7/10 on the unversioned endpoint, triggering freezes, disclosure, re-baselining, and a later attestation gate. GPT-4.1's sentinel cells remained 10/10. Separately, Claude Haiku failed the registered behavioral entry gate and was replaced through a sealed amendment. This is a procedural record, not a behavioral-effect estimate. Together the cases show that model identity and endpoint availability do not by themselves establish a stable, usable behavioral subject.

S5.3 Prospective replication

A separately authorized prospective replication could preselect one target or a small candidate family and preregister the complete familywise procedure: candidate set, episode-level unit, interiority gate, maximum statistic, decision threshold, and sample size. For the composition estimand, it will either widen the preregistered prompt family or prospectively sample from a declared persona generator rather than reuse the fixed sixteen prompts as if they represented a population. For the response estimand, the episode allocation will be chosen by simulating the exact registered gate and candidate family until the minimum attainable familywise p-value is below the declared threshold and power meets a prespecified target under the declared alternative. The design should also include a format-matched neutral prefix, a continuation-probability × wording factorial, and a structurally equivalent non-PD label-conflict control.

S5.4 Research record

The complete correction ledger, sealed discussion text, dead-predictions ledger, reviewer-role disclosures, and mechanical disposition matrices are maintained in the public Review Archive and Submission Analyses directories. Persona provenance is documented in the Persona Table; scholarly positioning is documented

in the Literature Map and Novelty Relationships documents. Historical artifacts are never silently rewritten; current interpretations are linked to the versions they amend.

S6. Sealed discussion in full

The source `docs/paper/discussion-branches.md` is preserved unchanged (SHA256 1f1d7de9 c54811962416a43bc5eed05de7fdd99618d39c88e5a8ec2cde9f0356). All four branches below were written before data; they are not four results. The historical selection was **Branch 2**, P5-2 **persona-dominant**, selecting its **not-task-dominant** paragraph. Branches 1/3/4 and other variants were unselected counterfactual prose, not findings. Source: docs/phase5-close/branch-selection.md and docs/phase5/final-report.md.

Sealed claim: P5-1 corner mixture; **Historical outcome:** P5-1a supported 3/32<.10; P5-1b matched heuristic thresholds; **Current scientific interpretation:** Method-sensitive: Jeffreys 5/32 fails. Human SD comparison is nonmatched/shape-blind; no prior-robust corner-purity or human-equivalence conclusion.

Sealed claim: P5-2 task-text dominance; **Historical outcome:** Prediction failed; mechanical persona-dominant variant selected; **Current scientific interpretation:** Pooled classification remains historical; prior/target sensitive and carried by confounded swap cell, not a general dominance hierarchy.

Sealed claim: P5-3 zero personas pass; **Historical outcome:** Prediction failed; 16/16 pass clause(b), p13 also passed uncontrolled clause(a); **Current scientific interpretation:** Choice-level clause(b) remains; latent agency/human-likeness interpretation unsupported. P13 capability interpretation withdrawn without prospective family control; replication target only.

Sealed claim: P5-4 no temperature refutation; **Historical outcome:** Not refuted on registered sweep; **Current scientific interpretation:** Bounded non-refutation, not zero temperature effect; entropy secondary and unmatched-composition issues separate. High-T slope branch is not a demonstrated finding.

The following historical text is reproduced for provenance. It is superseded where the main article and S3 withdraw its capability interpretation. Its header and prediction errors are part of the sealed record.

Paper one — pre-committed discussion branches (Phase 5)

> **STATUS: PROPOSED — UNSEALED.** This file seals (hash-anchored with registry v4) BEFORE any Phase 5 dispatch. After seal it is immutable: the published paper's discussion section must be byte-diffable against the branch the registered verdicts select. The author does not get a vote after data exists. The completeness linter (check C5) treats a registered verdict combination without a written branch as a seal failure.

Verdict axes and branch-selection rule (registered)

Three registered verdict axes select the branch:

- **A** — P5-1 corner mixture: `supported` / `not-supported`
- **B** — P5-3 interior persona existence: `zero` / `at-least-one`
- **C** — P5-4 temperature refutation (interior fraction rises with T, or a payoff-graded response appears at high T): `no` / `yes`

Selection rule, applied in this order: if **B = at-least-one** → Branch 2. Else if **C = yes** → Branch 3. Else if **A = supported** → Branch 1. Else → Branch 4.

P5-2 (surface-cue dominance hierarchy) does not select the branch; every branch contains both pre-written P5-2 variant paragraphs (task-dominant / not task-dominant) and the published text keeps exactly the variant the P5-2 verdict selects.

Combination table (exhaustive; linter-checked)

A (P5-1): supported; **B (P5-3):** zero; **C (P5-4):** no; **Branch:** Branch 1 — corners everywhere

A (P5-1): supported; **B (P5-3):** zero; **C (P5-4):** yes; **Branch:** Branch 3 — temperature-graded

A (P5-1): supported; **B (P5-3):** at-least-one; **C (P5-4):** no; **Branch:** Branch 2 — an interior persona exists

A (P5-1): supported; **B (P5-3):** at-least-one; **C (P5-4):** yes; **Branch:** Branch 2 — an interior persona exists

A (P5-1): not-supported; **B (P5-3):** zero; **C (P5-4):** no; **Branch:** Branch 4 — mixed

A (P5-1): not-supported; **B (P5-3):** zero; **C (P5-4):** yes; **Branch:** Branch 3 — temperature-graded

A (P5-1): not-supported; **B (P5-3):** at-least-one; **C (P5-4):** no; **Branch:** Branch 2 — an interior persona exists

A (P5-1): not-supported; **B (P5-3):** at-least-one; **C (P5-4):** yes; **Branch:** Branch 2 — an interior persona exists

---

Branch 1 — corners everywhere

The question Phase 5 asked was whether any conditioning of the subject — content-side, through persona system prompts, or decoding-side, through temperature — could produce a human-like interior behavioral population. The answer is no. Sixteen personas generated by a seeded procedure from crossed trait factors, evaluated across the program's main instruments at three temperatures, produced pools that are mixtures of the same corners the bare subject occupies: the fraction of persona-cells with interior episode-level cooperation stayed below the registered threshold, and between-persona variance fell short of the published human within-condition panels cited in the sealed predicates. Personas relocate the corners and decorate them with demographic texture; they do not populate the interior between them.

This completes the program's central claim in its strongest available form. Phases 3 and 4 established that the bare subject's behavior is driven by local semantic-textual content rather than payoffs or display mechanics. Phase 5 shows the result is not an artifact of the bare subject's blankness: no member of a registered persona population passed the two-sided assay gate with a positive discount-factor slope, and none refused the dominated mislabeled option at the registered rate. Incentive sensitivity — the signature of economic agency that every human panel in the cited literature exhibits at the individual level — did not appear anywhere in the conditioned population, at any temperature.

P5-2 variant (task-dominant): The dominance hierarchy sharpens the mechanism. Where a persona's registered leaning conflicted with the known task-text switches, the task text won at the registered margin: local semantic content beats global framing one layer up, exactly as the Phase 4 thesis predicted. The system prompt is weaker than the words in front of the model.

P5-2 variant (not task-dominant): The dominance hierarchy did not resolve at the registered margin: in conflict cells, persona leaning and task-text switches produced mixed control of choice. This bounds the Phase 4 thesis — local semantic content dominates the bare subject, but a global persona frame can contest it — without disturbing the corner-mixture result, which held regardless of which cue won.

Temperature behaved as registered: raised T manufactured variance — symmetric flip-noise around the corners and elevated invalid rates — without manufacturing behavior. The interior fraction did not increase, and no payoff-graded response appeared at any temperature. For simulation practice the implication is direct and negative: persona-conditioning and decoding heat, the two levers practitioners actually pull to make synthetic subjects "diverse," produce populations that are diverse in costume and uniform in mechanism. Synthetic-player pools built this way cannot substitute for human panels in any design where interior heterogeneity or incentive sensitivity carries the identification.

Branch 2 — an interior persona exists

The headline of Phase 5 is an existence result the program registered against itself: at least one persona in the sealed sixteen passed the two-sided assay gate and showed the registered signature of incentive sensitivity. The author's registered prediction — that none would — is refuted, and the refutation is the finding. Human-likeness, absent in the bare subject throughout Phases 3 and 4, can be relocated into the conditioning layer: some region of persona space induces interior behavior with a positive discount-factor slope (or registered-rate refusal of the dominated mislabeled option), where the bare subject shows corners and surface-cue tracking.

Which persona(s) passed, and their registered trait factors, are reported in the results section exactly as adjudicated; nothing in this pre-committed text depends on which cell cleared the gate. What the existence result changes is the interpretation of the program's earlier phases: the corner behavior of Phases 3–4 characterizes the bare subject, not the model's capability envelope. The capability was recoverable by content-side conditioning that contains no game-relevant instruction — trait words only, under the sealed banned-content guard.

P5-2 variant (task-dominant): Even so, the dominance hierarchy held where tested: in registered conflict cells, task-text switches beat persona leaning at the registered margin. Interior behavior and surface-cue dominance coexist — conditioning moved the operating point without transferring control from local semantic content to the global frame.

P5-2 variant (not task-dominant): Consistent with the existence result, the dominance hierarchy did not resolve for the task side: persona framing contested or beat the task-text switches in registered conflict cells. The two results together indicate that the system-prompt layer is not inert decoration but a genuine second channel of behavioral control.

The scope of the claim is deliberately narrow. One (or few) passing persona(s) out of sixteen is not a human-like population; the population predicates (P5-1) are reported alongside, and the between-persona variance comparison against the cited human panels bounds how much of the interior the pool actually covers. Whether the passing region of persona space is stable, extensible, or mechanistically interpretable is future work — and under the sealed scope rule, it stays there: no new arms were added, and the program ends with paper one.

Branch 3 — temperature-graded

The registered temperature prediction is refuted, and that refutation is the finding of Phase 5. Raising decoding temperature did not merely add symmetric flip-noise around the corners: the interior fraction rose with T (or a payoff-graded response appeared at high temperature where it was absent at the program constant), meeting the registered refutation predicate — structure of the kind quantal-response models describe, produced by the decoding layer rather than by content-side conditioning.

This inverts the program's working picture in one specific way. Through Phases 3 and 4, behavior at the program's constant temperature was corner behavior, switched by local semantic content; the natural reading was that the model has no interior to expose. The temperature result shows the interior exists in the decoding distribution and is suppressed, not absent, at low temperature. Sampling heat, not persona content, is what exposes it: persona pools at the program constant remained corner mixtures under the P5-1 predicates (as reported alongside), while the registered temperature sweep moved cells into the interior.

P5-2 variant (task-dominant): The dominance hierarchy still resolved for the task side at the registered margin — local semantic switches beat persona leaning in conflict cells — so the graded structure arrives from the decoder while control of direction stays with the task text.

P5-2 variant (not task-dominant): The dominance hierarchy did not resolve at the registered margin, so Phase 5 leaves control of direction mixed between persona frame and task text even as the decoder supplies graded structure.

Two registered cautions bound the interpretation. First, invalid-trial rate by temperature was a registered secondary outcome precisely because parse degradation can masquerade as behavioral change; the reported decomposition separates the two, and the refutation predicate was evaluated on the behavioral component alone. Second, QRE-like is a description of the observed grading, not an identification of a utility structure; nothing in this design estimates a rationality parameter. What can be said is exactly this: a registered, adversarial prediction that temperature only manufactures variance was put on the record and the data refused it.

Branch 4 — mixed

Phase 5 returns a mixed verdict, and under this program's rules a mixed verdict is reported as exactly that — no narrative rescue in either direction. The corner-mixture predicate (P5-1) did not reach its registered support threshold, but no persona passed the individual incentive- sensitivity gate (P5-3), and temperature produced no registered refutation (P5-4): the population is neither corner-pure enough to support the strong mixture claim nor human-like on any registered individual or decoding predicate.

Concretely, this is the outcome in which persona conditioning smears behavior — some persona-cells drift off the exact corners, between-persona variance may approach segments of the human panels — without any cell exhibiting the registered signatures that would make the drift human-like: no two-sided interior operating point with a positive discount-factor slope, no registered-rate refusal of the dominated option, no temperature-graded structure. Decorated corners with blur is still the most accurate summary the registered predicates permit.

P5-2 variant (task-dominant): The hierarchy result stands on its own: task-text switches beat persona leaning at the registered margin in conflict cells, extending the Phase 4 thesis one layer up even where the population predicates went mixed.

P5-2 variant (not task-dominant): The hierarchy result is also unresolved: neither cue controlled choice at the registered margin in conflict cells. Phase 5 then closes with both its population-level and its control-level questions returning mixed answers.

For the paper's conclusion the mixed branch carries a specific obligation: every predicate is reported with its registered threshold and its observed value, the misses in both directions are enumerated, and the temptation to round the result toward either headline is refused mechanically — this text was written before the data existed, and the numbers that follow it in the published version are whatever the adjudicators produced. What survives untouched from the earlier phases is the Phase 3–4 record itself: the bare subject's corner behavior and surface-cue sensitivity were established under their own sealed predicates and are not revised by Phase 5's mixed population result. Under the sealed scope rule, resolving the blur — different persona geometry, richer instruments, other decoders — is future work, and paper one ends here.

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